

**MATH 2101 LINEAR ALGEBRA I, FALL 2025 – OUTLINE OF
SOLUTIONS FOR ASSIGNMENT 2**

Due date: 22 September (Monday), 2025, 10:00 pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
 - All solutions have to be turned in HKU moodle in the format of *PDF file*.
 - Please include your *Name, UID, Faculty, Major (if declared)* in your solution.
1. Let A be an invertible 5×5 matrix and let B be an invertible 3×3 matrix. Let C be a 5×3 matrix. Consider the following block matrix:

$$D = \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix}.$$

Determine if D is invertible. If D is invertible, find D^{-1} in terms of A^{-1} , B^{-1} , A , B , C . If D is not always invertible, give an example.

Sol: We claim that the following 8×8 matrix

$$E = \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix}$$

is the inverse of D . In particular, D is invertible. Indeed, we will verify that $ED = DE = I_8$. Using the multiplication rule for block matrices, and the fact that A and B are invertible, one has

$$\begin{aligned} ED &= \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix} \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix} \\ &= \begin{pmatrix} A^{-1}A & A^{-1}C - A^{-1}CB^{-1}B \\ 0_{3 \times 5} & B^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} I_{5 \times 5} & 0_{5 \times 3} \\ 0_{3 \times 5} & I_{3 \times 3} \end{pmatrix} = I_8. \end{aligned}$$

Similarly, one has

$$\begin{aligned} DE &= \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix} \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix} \\ &= \begin{pmatrix} AA^{-1} & -AA^{-1}CB^{-1} + CB^{-1} \\ 0_{3 \times 5} & BB^{-1} \end{pmatrix} \\ &= \begin{pmatrix} I_{5 \times 5} & 0_{5 \times 3} \\ 0_{3 \times 5} & I_{3 \times 3} \end{pmatrix} = I_8. \end{aligned}$$

Therefore, we conclude that $D^{-1} = E$, which can be expressed using A^{-1} , B^{-1} , A , B and C as above.

Remark. One way to obtain E is to assume it is of the form

$$E = \begin{pmatrix} X & Y \\ Z & W \end{pmatrix},$$

which is of the same block structure as D , i.e., X and A have the same block size, Y and C have the same block size etc. Upon stipulating that $ED = I_8$, we can obtain E in the form above. However, one still has to use the definition of matrix inverse to show that $E = D^{-1}$.

2. Let v be an $n \times 1$ matrix. Suppose $n \geq 2$. Use properties of determinants to prove that $\det(vv^T) = 0$.

Sol: Let $v = (a_1, a_2, \dots, a_n)^T$. Then

$$\begin{aligned} \det(vv^T) &= \det \begin{pmatrix} a_1 v^T \\ a_2 v^T \\ \vdots \\ a_n v^T \end{pmatrix} \\ &= a_1 a_2 \dots a_n \det \begin{pmatrix} v^T \\ v^T \\ \vdots \\ v^T \end{pmatrix} \\ &= 0, \end{aligned}$$

where the first equality follows from matrix multiplication, the second equality follows from the scalar rule in computing determinants and the third equality follows from that there are two identical rows in the matrix.